LEVEL 3 · ENTERPRISE / REMOTE ACCESS (DMZ)

FW-01 Industrial firewall
DMZ · read-only data mirror
Only Level 2 ↔ Level 3 conduit

Remote monitoring
Ministry / O&M
VPN + MFA (TBC by client)

VPN

LEVEL 2 · SUPERVISORY – SCADA LAN, VLAN 10, 10.10.10.0/24 (CONTROL ROOM)

HMI-01
Operator station 1
HMIP6 · MSE runtime 4K

HMI-02
Operator station 2
HMIP6 · MSE runtime 4K

SRV-01
SCADA server / historian
13-month storage (TBC)

ENG-01
Engineering station
Control Expert · MSE BT

GPS-01
GPS master clock
NTP / PTP time source

SW1 · MCSESM083F23F1
Managed switch 8TX, control room
VLAN 10 / 20, RSTP, SNMPv3

SCADA LAN – VLAN 10 (via SW1)

LEVEL 1 · CONTROL – M580 PLANT CONTROLLER, VLAN 20, 10.10.20.0/24

NOC0311 #1 / #2

MAIN RACK – BMEXBP0800 (control room)
BMXCPS3020 power supply 24–48 V DC
BMEP582040 M580 CPU – plant controller / PPC
BMENOC0311 #1 – SCADA LAN (Modbus TCP server)
BMENOC0311 #2 – second SCADA LAN path
BMENOC0321 – device network scanner, IP forwarding
BMXDDI3202K ×2 · BMXDDO3202K ×2
Spare BMEP582040 held as cold spare (see note 3)

RIO

RACK 2 – BMEXBP1200 (11 kV GIS room)
BMXCPS3020 power supply
BMECRA31210 Ethernet drop adapter
BMXDDI3202K ×3 – breaker status, trips, fire
BMXDDO3202K ×2 – breaker / DG / BESS commands
BMXNOM0200 – RS485 ×2 (DG controllers)
MCSESM083F23F1 SW2 mounted in this panel

SW2 · MCSESM083F23F1
Device network switch, VLAN 30
Fibre ring to field switches

NOC0321 → SW2 (device network, IP forwarding)

Fibre ring via field switches – see sheet 002

RS485 to DGC-1/2

LEVEL 0 · FIELD DEVICES – DEVICE NETWORK, VLAN 30, 10.10.30.0/24, AND SERIAL BUSES

CONTROL ROOM & 11 kV GIS ROOM

P5M30-AH1 … AH8
8 × Easergy P5M30, RSTP ring
Protection, measurement, events

GIS AH1–AH8 hard-wired
VCB / switch positions, trips, SF6,
L/R, VPI → Rack 2 DI; CLOSE/OPEN DO

NECR (AH8)
Earthing TX trip, NER 63.5 Ω
temperature via P5M30-AH8

UPS-01 + 110 V DC charger
Control room UPS, GIS DC system
Modbus TCP card

FACP-01 fire panel
GIS room, control room zones
Volt-free contacts → Rack 2

AUX panel (AH5)
XRNT-3A fuse, 100 kVA TX, 200 A
Incomer status → Rack 2

BESS CONTAINER

BESS-EMS (vendor)
Energy management controller
Modbus TCP server to M580

PCS-1 / PCS-2
Grid-forming PCS (qty TBC)
via EMS

BMS racks 1–8
3.34 MWh LFP 416S8P
via EMS

PM5320-BESS
BESS LV metering 0.69 kV

BESS ACB / fire / ESD
ACB 1000 A, fire, off-gas,
E-stop → Rack 2 DI / DO

DIESEL GENERATOR AREA

DGC-1 (DG1 550 kVA)
Genset controller
RS485 → BMXNOM0200 port 1

DGC-2 (DG2 660 kVA)
Genset controller
same bus, ID 2

PM5320-DG
DG LV bus metering

ACB Q2 / Q3, start/stop
Hard-wired → Rack 2

PV INVERTER STATION & ARRAY

NTS-PV1 … PV4
4 Edge I/O NTS islands with
NTSFMB0120K serial master each

INV NB01 … NB04
Sungrow SG350HX 350 kW
RS485 → NTSFMB0120K

PAS600-PV1 … PV4
Panel servers + TH110 / CL110
enclosure temperature / humidity

WS-01 weather station
Meteocontrol blue'Log XM 5000
SMP12-A, WindSonic, Tamb, Tmod

PM5320-PV
PV LV metering 0.8 kV

PV ACB 1250 A
Status / command → Rack 2

PRESIDENT TX STATION

NTS-LVP1 / LVP2
RMU, TX, LVDB I/O

PAS600-LVP1 / 2
TH110 / CL110

PM5320-LVP
LVDB metering

RMU ST260E
contacts → NTS

UPS-02 2 kVA
RMU UPS card

PM TX STATION

NTS-LVM1
RMU, TX, LVDB

PAS600-LVM1
TH110 / CL110

PM5320-LVM
LVDB metering

RMU ST260E
contacts → NTS

UPS-03 2 kVA
RMU UPS card

LEGEND

- EtherNet/IP (M580 RIO / DIO)
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- RS485 – Modbus RTU
- Hard-wired I/O (DI / DO)
- Vendor-internal link (not SCADA scope)
- Device; stripe = interface type to SCADA
- Network level / location boundary

NOTES

1. Architecture follows the Schneider Electric offer (Villa Somalia v1) and drawing BFE/PR/26/01. Device names are used unchanged on sheets 002–005, the HMI and the I/O list.
2. The M580 is the plant controller (PPC): it dispatches BESS, PV and DG setpoints. Protection relays trip independently of SCADA.
3. BMEP582040 is not hot-standby capable. The second CPU in the BOM is shown as a cold spare. Change to an M580 hot-standby CPU if redundancy is required (TBC).
4. SCADA server, historian, engineering station, GPS clock and firewall are not in the BOM and must be added (TBC).
5. Field switches at remote locations, BESS EMS interface, DG controller interface and FACP interface are not in the BOM (TBC).
6. Only the firewall conduit connects Level 2 to Level 3. Remote access is subject to client approval.
7. Equipment locations and cable distances to be confirmed against the general layout.

REV	DESCRIPTION	DATE
A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

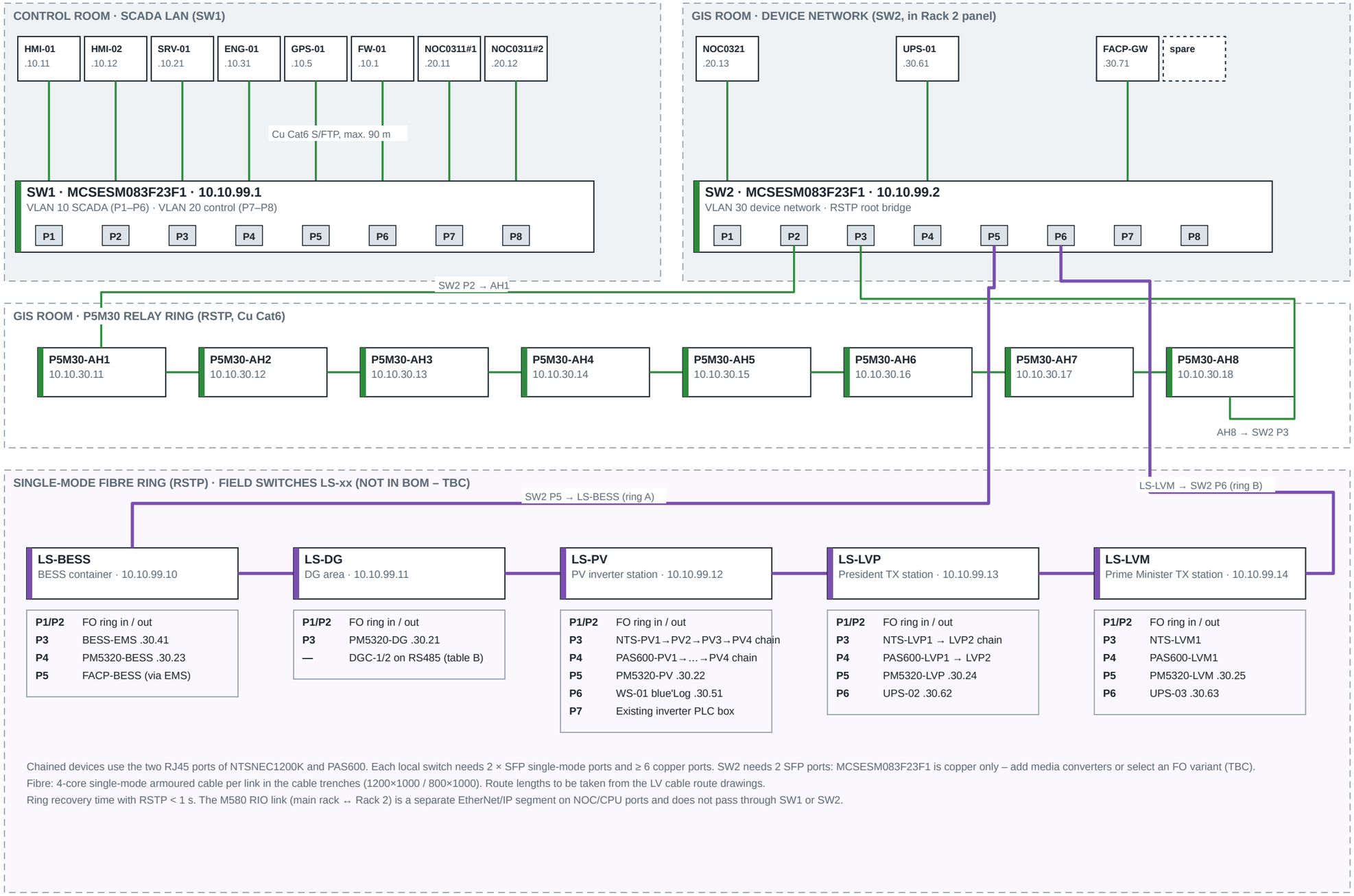
CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
SCADA ARCHITECTURE
System overview by level and location

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-001** **REV A**



P1/P2	FO ring in / out
P3	BESS-EMS .30.41
P4	PM5320-BESS .30.23
P5	FACP-BESS (via EMS)

P1/P2	FO ring in / out
P3	PM5320-DG .30.21
—	DGC-1/2 on RS485 (table B)

P1/P2	FO ring in / out
P3	NTS-PV1→PV2→PV3→PV4 chain
P4	PAS600-PV1→…→PV4 chain
P5	PM5320-PV .30.22
P6	WS-01 blue'Log .30.51
P7	Existing inverter PLC box

P1/P2	FO ring in / out
P3	NTS-LVP1 → LVP2 chain
P4	PAS600-LVP1 → LVP2
P5	PM5320-LVP .30.24
P6	UPS-02 .30.62

P1/P2	FO ring in / out
P3	NTS-LVM1
P4	PAS600-LVM1
P5	PM5320-LVM .30.25
P6	UPS-03 .30.63

Chained devices use the two RJ45 ports of NTSNEC1200K and PAS600. Each local switch needs 2 × SFP single-mode ports and ≥ 6 copper ports. SW2 needs 2 SFP ports: MCSESM083F23F1 is copper only – add media converters or select an FO variant (TBC).
Fibre: 4-core single-mode armoured cable per link in the cable trenches (1200×1000 / 800×1000). Route lengths to be taken from the LV cable route drawings.
Ring recovery time with RSTP < 1 s. The M580 RIO link (main rack ↔ Rack 2) is a separate EtherNet/IP segment on NOC/CPU ports and does not pass through SW1 or SW2.

TABLE A – VLAN AND IP ADDRESS PLAN

VLAN	Subnet	Use	Gateway / routing
10	10.10.10.0/24	SCADA LAN (HMI, server, ENG)	FW-01 .1 (to DMZ only)
20	10.10.20.0/24	M580 control (CPU, NOC, CRA)	None
30	10.10.30.0/24	Field devices	BMENOC0321 IP forwarding
99	10.10.99.0/24	Switch management	ENG-01 only

Device	IP address	Remark
FW-01 / GPS-01	10.10.10.1 / .5	NTP server for all devices
HMI-01 / HMI-02 / SRV-01 / ENG-01	.10.11 / .12 / .21 / .31	Static addresses
M580 CPU / NOC0311 #1 / #2 / NOC0321	10.10.20.10 / .11 / .12 / .13	NOC0321 also 10.10.30.1
BMECRA31210 (Rack 2)	10.10.20.20	RIO drop
NTS-PV1…PV4 / LVP1, LVP2 / LVM1	10.10.30.101–104 / 105, 106 / 107	EtherNet/IP adapters
P5M30-AH1 … AH8	10.10.30.11 … 18	Modbus TCP, unit ID 1
PM5320 DG / PV / BESS / LVP / LVM	10.10.30.21 … 25	Modbus TCP
PAS600-PV1…4 / LVP1, 2 / LVM1	10.10.30.31 … 37	Modbus TCP gateway for sensors
BESS-EMS / WS-01 / UPS-01…03 / FACP-GW	.30.41 / .51 / .61–.63 / .71	Modbus TCP

TABLE B – RS485 MODBUS RTU BUSES

Bus	Devices (Modbus ID)	Settings	Remark
NOM0200 port 1	DGC-1 (1), DGC-2 (2)	19200 8N1	Shielded twisted pair, 120 Ω ends
NOM0200 port 2	Spare	—	Reserved
NTSFMB0120K PV1	INV NB01 (1)	9600 8N1	Point-to-point, see note 4
NTSFMB0120K PV2	INV NB02 (1)	9600 8N1	Point-to-point
NTSFMB0120K PV3	INV NB03 (1)	9600 8N1	Point-to-point
NTSFMB0120K PV4	INV NB04 (1)	9600 8N1	Point-to-point
NTSFMB0120K LV	Spare	—	Future meters

TABLE C – CABLE TYPES

Type	Specification
Ethernet Cu	Cat6 S/FTP industrial, RJ45 shielded, ≤ 90 m
Fibre	Single-mode OS2, 4-core armoured, LC duplex, 1310 nm
RS485	2-pair shielded twisted pair 120 Ω (e.g. Belden 3106A), shield earthed at master end
Hard-wired I/O	Multicore 1.0 mm² screened, 24 V DC, separate from power cables

LEGEND

- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- RS485 – Modbus RTU
- EtherNet/IP (M580 RIO / DIO)
- Switch port Pn

NOTES

1. Read with sheet 001. Addresses are proposals for approval; final values to be fixed at FAT.
2. All switches managed: unused ports shut down, port security (MAC binding), SNMPv3, syslog to SRV-01, RSTP with SW2 as root bridge.
3. IP forwarding on BMENOC0321 is restricted to SCADA ↔ device traffic; no other routing between VLANs except via FW-01.
4. Communication schematic BFE/PR/26/01 shows the 4 inverters daisy-chained to one PLC box. The offer provides one NTSFMB0120K per PV area, so point-to-point links are proposed for faster polling and fault isolation (TBC).
5. Field switches LS-xx, SFP ports on SW2 and fibre cables are additions to the BOM (TBC).
6. All devices take time from GPS-01 via NTP; P5M30 event time-stamps at 1 ms.

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A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

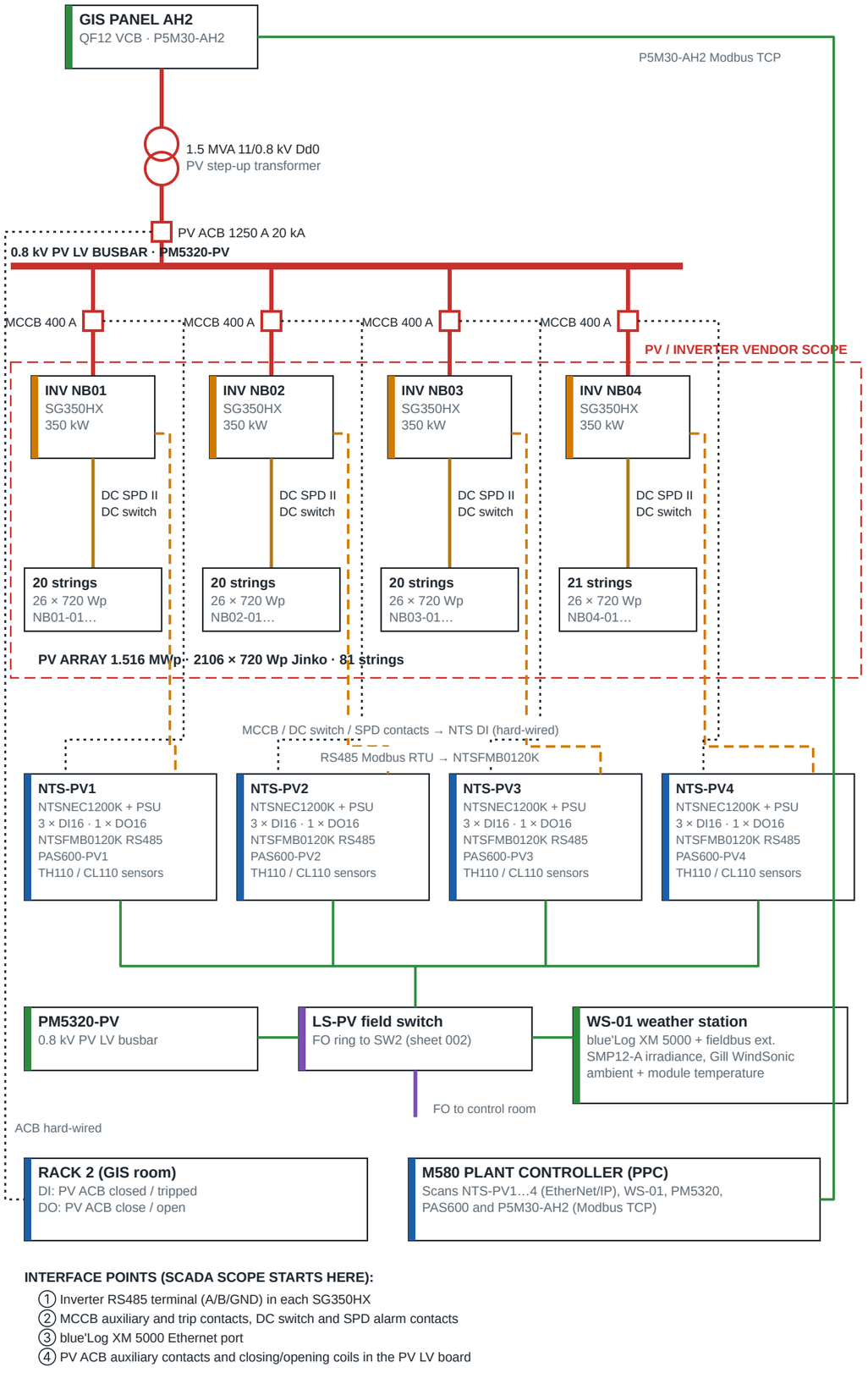
PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
COMMUNICATION NETWORK
Topology, ports, IP plan and serial buses

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-002** **REV A**

POWER AND SIGNAL INTERFACE – PV PLANT



INTERFACE POINTS (SCADA SCOPE STARTS HERE):

① Inverter RS485 terminal (A/B/GND) in each SG350HX
② MCCB auxiliary and trip contacts, DC switch and SPD alarm contacts
③ blue'Log XM 5000 Ethernet port
④ PV ACB auxiliary contacts and closing/opening coils in the PV LV board

TABLE A – SIGNAL EXCHANGE (PER INVERTER UNLESS STATED)

Dir.	Signal	Type	Source	Rate
IN	AC P, Q, U, I, f; DC P, U	AI comm	SG350HX	2 s
IN	Energy today / total	AI comm	SG350HX	60 s
IN	Internal temp., insulation resistance	AI comm	SG350HX	10 s
IN	Operating state, fault code	AI comm	SG350HX	2 s
IN	String currents (20 / 21 per inverter)	AI comm	SG350HX	10 s
OUT	Active power limit (% of 350 kW)	AO comm	PPC	on change
OUT	Reactive power / PF setpoint	AO comm	PPC	on change
OUT	Start / stop	AO comm	Operator	command
IN	MCCB 400 A closed / tripped	DI	NTS-PVn	20 ms
IN	DC switch closed, AC/DC SPD failed	DI	NTS-PVn	20 ms
IN	Enclosure temperature / humidity	AI comm	PAS600-PVn	60 s
IN	POA irradiance, Tamb, Tmod, wind	AI comm	WS-01 (plant)	5 s
IN	PV LV P, Q, U, I, PF, energy, THD	AI comm	PM5320-PV	1 s
IN	PV ACB closed / tripped	DI	Rack 2	20 ms
OUT	PV ACB close / open	DO	Rack 2	command
IN	AH2 protection, measurement, events	DI/AI comm	P5M30-AH2	1 s

Signal count (I/O list, area PV): 18 DI · 1 DO · 155 AI comm · 12 AO comm · plus weather station 7 and AH2 relay points.

TABLE B – CONTROL FUNCTIONS AND BEHAVIOUR

Function	Description
Curtailment	PPC writes P limit to all inverters when BESS reaches charge limit or SOC ≥ 97 %.
	Ramp rate 10 %/s (TBC with BESS vendor).
Reactive power	Inverters at PF 1.0 by default; Q setpoint available to PPC.
Close permissive AH2	Bus live (BESS or DG forming the grid). PV cannot energise the bus.
Loss of grid former	Inverters trip on their own V / f protection (anti-islanding).
Comms loss PLC ↔ inverter	Inverter comms watchdog 30 s → fallback P limit (vendor setting, TBC).
	SCADA raises P2 alarm, values shown with bad quality.
String monitoring	String current vs. array average: warn at −10 %, alarm at −30 %.
Performance ratio	Calculated in SCADA from PV energy and WS-01 irradiation.

TABLE C – INTERFACE RESPONSIBILITY

Item	PV / inverter vendor	SCADA contractor
Modbus register map SG350HX	Supply	Implement
RS485 cable inverter → NTS	Terminate at inverter	Supply, lay, test
Inverter Modbus ID / baud setting	Set	Verify
Auxiliary contacts MCCB / SPD	Wire to terminal strip	Wire to NTS
Weather station sensors & logger	Supply, install	Integrate (Modbus TCP)
Curtailment test at SAT	Witness	Perform

LEGEND

- Power circuit
- DC circuit
- RS485 – Modbus RTU
- EtherNet/IP (M580 RIO / DIO)
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- Hard-wired I/O (DI / DO)
- Vendor scope boundary

NOTES

1. Ratings from Overall SLD and inverter wiring diagram BFE/PR/26/01.
2. One NTS island and one PAS600 per PV area as per the offer (Inverter & DC combiner 1–4).
3. The BOM line PVSCL60E (Schneider string inverter) does not apply: the plant uses Sungrow SG350HX. Remove from BOM.
4. Modbus register map of SG350HX to be obtained from Sungrow before FAT.
5. Signal names follow the I/O list (area PV, tags VS_PV_INV_NB0x_…).

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MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
PV INTERFACE
Signal exchange, control and scope boundary

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
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APPROVED		SIGN
DRW NO.	BFE/PR/26/01-SCADA-003	REV A

POWER AND SIGNAL INTERFACE – BATTERY ENERGY STORAGE

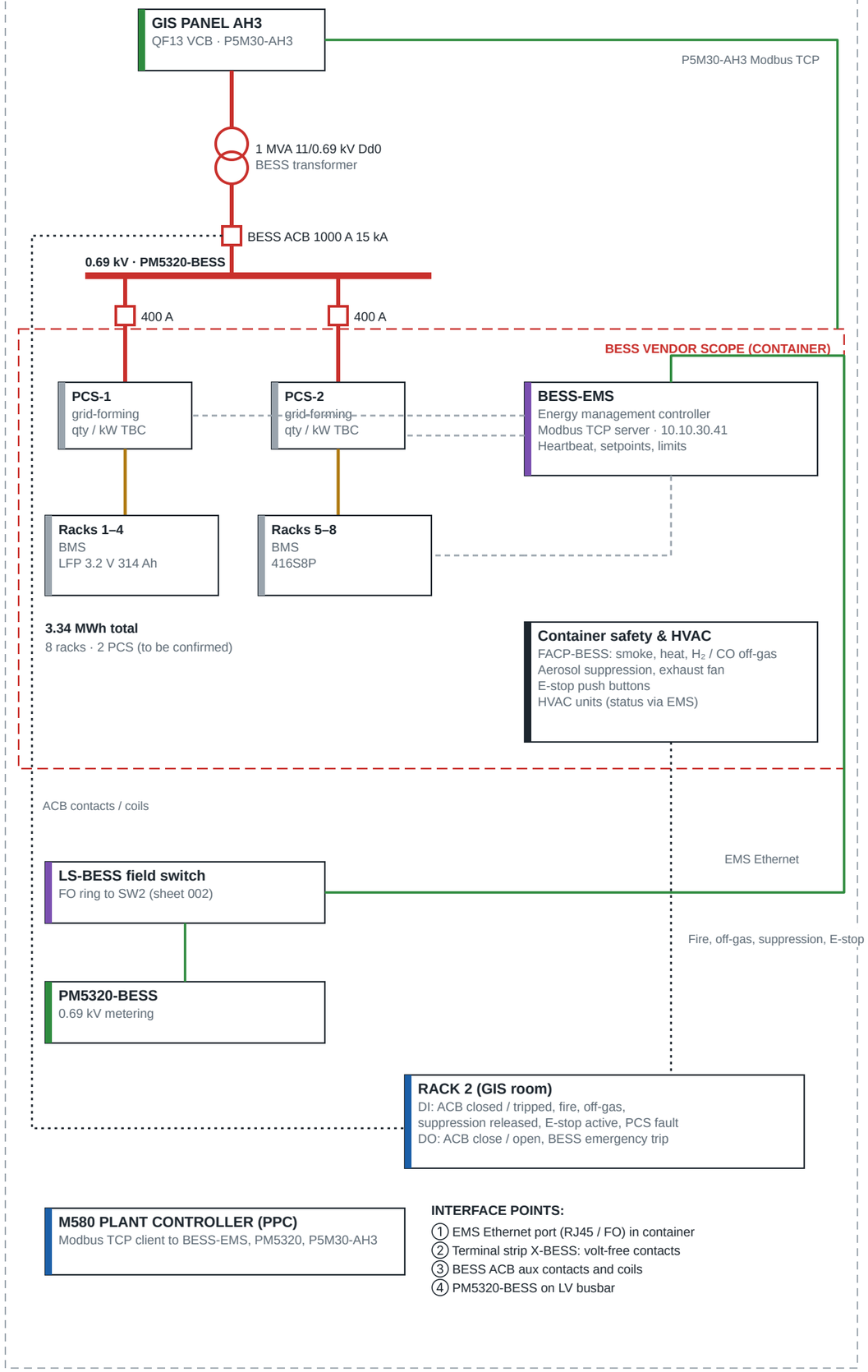


TABLE A – SIGNAL EXCHANGE SCADA ↔ BESS

Dir.	Signal	Type	Path	Rate
IN	SOC, SOH, energy stored, charged / discharged	AI comm	EMS	1 s
IN	Max charge / discharge power allowed	AI comm	EMS	200 ms
IN	P, Q, U, f per PCS; DC U, I	AI comm	EMS	200 ms
IN	Rack V, I, SOC, max temp, cell V max/min	AI comm	EMS	5 s
IN	PCS running, grid-forming, fault, warning	DI comm	EMS	200 ms
IN	HVAC temp / humidity, container status	AI comm	EMS	10 s
IN	EMS heartbeat counter	AI comm	EMS	1 s
OUT	PLC heartbeat counter	AO comm	PPC	1 s
OUT	P / Q setpoint, mode GF / GFL, start/stop	AO comm	PPC	on change
OUT	SOC limits (min / max) for PPC dispatch	AO comm	PPC	on change
IN	ACB closed / tripped	DI	Rack 2	20 ms
IN	Fire alarm, off-gas alarm, suppression released	DI	Rack 2	20 ms
IN	E-stop active, PCS common fault (backup)	DI	Rack 2	20 ms
OUT	ACB close / open, BESS emergency trip	DO	Rack 2	command
IN	LV P, Q, U, I, PF, energy	AI comm	PM5320	1 s
IN	AH3 protection, measurement, events	DI/AI comm	P5M30-AH3	1 s

Signal count (I/O list, area BESS): 6 DI · 2 DO · 88 AI comm · 16 DI comm · 8 AO comm.

TABLE B – OPERATING MODES

Mode	V / f set by	PPC role
Normal (islanded)	PCS grid-forming	Charge limit, SOC window, PV curtailment
DG support	PCS grid-forming	DG base load, BESS charges from DG
Black start	PCS grid-forming	Sequenced breaker closing (sheet 005 / HMI)
BESS unavailable	DG controller	DG isochronous, PV limited to load
Maintenance	—	BESS ACB open, AH3 earthed, permit to work

TABLE C – FAILURE BEHAVIOUR

Event	Response
EMS loses PLC heartbeat > 10 s	PCS keeps grid-forming with local droop; setpoints frozen.
PLC loses EMS heartbeat > 10 s	P1 alarm; DG auto-start; PV limit held at last value.
Fire / off-gas alarm	Hard-wired trip: PCS stop, ACB open, HVAC recirc. off,
	exhaust on. Works without communication.
E-stop in container	PCS stop and ACB trip by vendor circuit; status to SCADA.
SOC below minimum	Load shedding by LVDB priority if no DG available.
AH3 protection trip	Bus dead unless DG running; black start from BESS.

TABLE D – INTERFACE RESPONSIBILITY

Item	BESS vendor	SCADA contractor
EMS Modbus register map & heartbeat	Supply	Implement
Terminal strip X-BESS (volt-free)	Supply, wire inside	Cable to Rack 2
Fire / suppression system	Supply, commission	Monitor only
PCS control modes & droop settings	Set, document	Verify at SAT
Fibre patch panel in container	Space & power	Supply, install

LEGEND

- Power circuit
- DC circuit
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- Hard-wired I/O (DI / DO)
- EtherNet/IP (M580 RIO / DIO)
- Vendor-internal link (not SCADA scope)
- Vendor scope boundary

NOTES

1. Ratings from Overall SLD BFE/PR/26/01: 3.34 MWh 416S8P, 1 MVA 11/0.69 kV, 2 × 400 A to the PCS.
2. PCS quantity, rating and the EMS make are to be confirmed by the BESS vendor.
3. SCADA never releases the fire suppression; it monitors only.
4. Hard-wired trips act independently of the PLC and of communication.
5. Signal names follow the I/O list (area BESS, tags VS_BESS_…).

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MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
BESS INTERFACE
Signal exchange, modes and scope boundary

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
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DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN
DRW NO.	**BFE/PR/26/01-SCADA-004**	**REV A**

SCADA OVERVIEW SCREEN – FOLLOWS OVERALL SLD BFE/PR/26/01 (VALUES ILLUSTRATIVE)

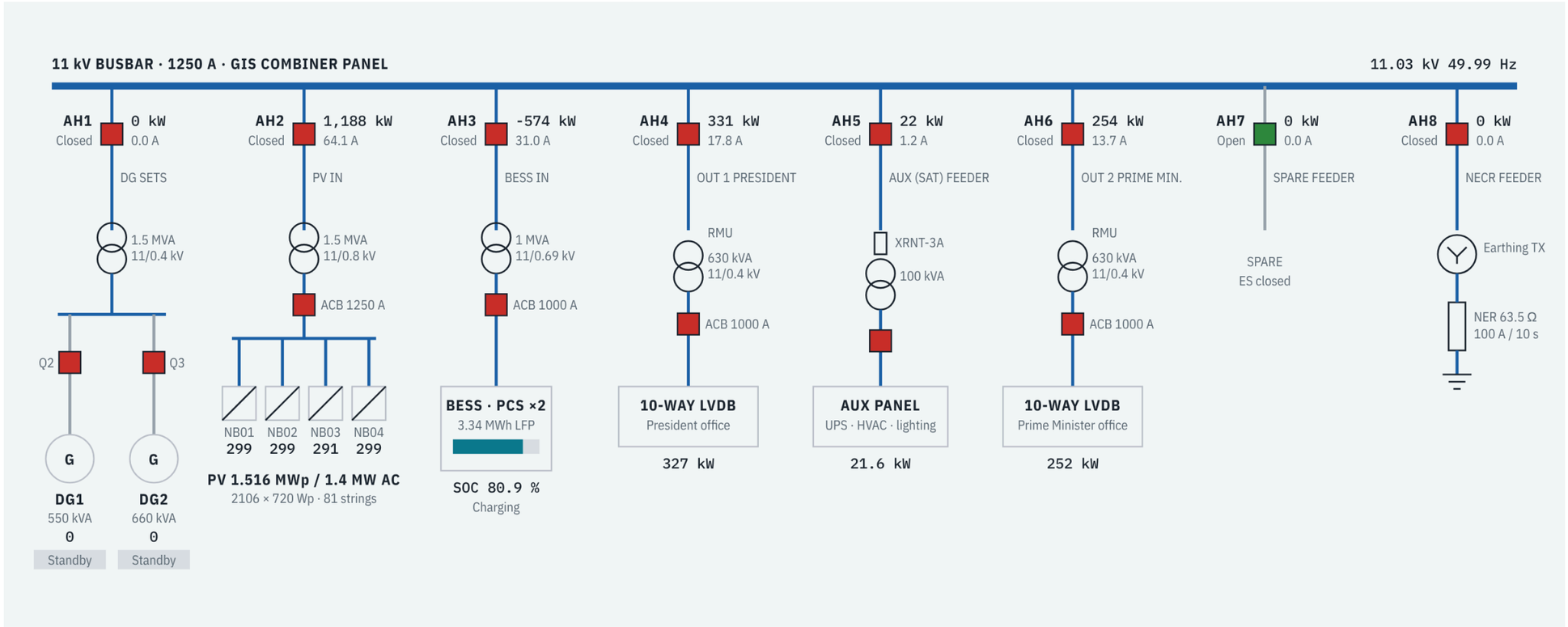


TABLE A – SYMBOL TO SIGNAL REFERENCE

Symbol	Status / measurement tags	Source
AH1…AH8 VCB	VS_GIS_AHn_QF_CLS / _OPN / _TRP	Rack 2 DI (hard-wired)
AH1…AH8 values	VS_GIS_AHn_P / _IA / _UAB	P5M30-AHn
AH1…AH8 command	VS_GIS_AHn_CMD_CLS / _CMD_OPN	Rack 2 DO, SBO
Earth switch	VS_GIS_AHn_QL_ES	Rack 2 DI
DG ACB Q2 / Q3	VS_DG_DGn_ACB_CLS / _ACB_TRP	Rack 2 DI
DG1 / DG2 block	VS_DG_DGn_P / _RUN / _AUTO	DGC-n (RS485)
PV ACB	VS_PV_LV_ACB_CLS / _ACB_TRP	Rack 2 DI
Inverter kW	VS_PV_INV_NB0n_PAC / _STATE	NTS-PVn (RS485)
BESS ACB	VS_BESS_LV_ACB_CLS / _ACB_TRP	Rack 2 DI
BESS block, SOC	VS_BESS_SYS_SOC / _P / PCSn_GF	BESS-EMS
LVDB incomer	VS_LV_LVDB_PRES/PM_INC_CLS	NTS-LVP / NTS-LVM
LVDB kW	VS_LV_LVDB_PRES/PM_P	PM5320-LVP / -LVM
Aux panel	VS_AUX_PNL_INC_CLS · VS_AUX_TX_FUSE	Rack 2 DI
NECR	VS_GIS_NECR_NER_TMP / _EFT_TRP	P5M30-AH8 / Rack 2
Bus kV / Hz	VS_GIS_AH3_UAB / _F (bus calc.)	P5M30, PLC

TABLE B – SYMBOL AND COLOUR CONVENTIONS

Element	Convention
Breaker closed	Filled red square, text "Closed"
Breaker open	Filled green square, text "Open"
Breaker tripped	Filled amber square, text "TRIPPED", P1 alarm
Energised conductor	Blue line
De-energised conductor	Grey line
Bad quality / comms lost	Value shown in magenta with "?" (to be added at detail design)
Alarm priority	P1 red · P2 amber · P3 blue, shown in alarm banner

TABLE C – OPERATOR INTERACTION

Action	Result
Click breaker	Faceplate: status, values, interlocks, select-before-operate
Click equipment	Opens detail screen (Generators, BESS, PV, Distribution, Aux)
Command	Select → operate within 10 s; PLC checks permissives
User level	Viewer: monitor only · Operator: commands · Engineer: setpoints
Logging	Every command and setpoint stored with user and time

SCREEN HIERARCHY

Level	Screens
1 – Plant	Overview SLD (this sheet)
2 – Area	11 kV GIS · Generators · BESS · PV & Weather · Distribution · UPS & Fire
3 – Function	Plant Control (PPC) · Alarms · Trends · Network · I/O List
4 – Faceplates	Breaker, DG, PCS, inverter, MCCB pop-ups

LEGEND

- Breaker closed
- Breaker open
- Breaker tripped
- Energised
- De-energised

NOTES

1. Screen captured from the HMI mock-up hmi/villa-somalia-scada.html. Values are simulated and for illustration only.
2. Layout follows the Overall SLD and 11 kV switchgear configuration of BFE/PR/26/01.
3. Tag names follow the I/O list docs/villa-somalia-io-list.csv.
4. Breaker status is from hard-wired auxiliary contacts and confirmed by the relay. Operators must verify locally before working on equipment.

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MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
SCADA OVERVIEW SLD
HMI overview screen and signal reference

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
DESIGNED		SIGN
DRAWN		SIGN
CHECKED		SIGN
APPROVED		SIGN

DRW NO. **BFE/PR/26/01-SCADA-005** **REV A**